



RS003 Week 4 — English Worksheet

Topic: A Three-Part Ship — Body + Nose + Thrusters · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week your ship grows from one rectangle into a **three-part craft**: a body rectangle, a pointed **nose** triangle on top, and two **thruster** blocks at the bottom. You build each part from variables, and you learn that **draw order** decides what sits in front.

Words to know: **polygon**, **triangle**, **palette**, **draw order**, **thruster**

1 · Predict 🧠

Read each piece of code. Before you run it, write what you think will happen.

```
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, HEIGHT - 80, 40, 50))
```

This is the ship body. Where does it sit on the screen and how big is it?


```
pygame.draw.polygon(screen, WHITE, [(CENTER_X, HEIGHT - 100), (CENTER_X - 20, HEIGHT - 80), (CENTER_X + 20, HEIGHT - 80)])
```

A polygon connects three points. One point is higher than the other two. What shape does it make, and where does it sit on the ship?


```
screen.fill(SPACE)
pygame.draw.rect(screen, CYAN, ...)    # body
pygame.draw.polygon(screen, WHITE, ...) # nose
pygame.draw.rect(screen, RED, ...)     # thruster
```

Things are drawn from first to last. Which one ends up on top — the space, the body, or the nose?

2 · Run It 🏃

🎯 Type the example, run it, and name the three parts

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

SPACE = (10, 12, 40)
CYAN = (80, 220, 255)
WHITE = (255, 255, 255)
RED = (230, 60, 60)

y = HEIGHT - 80

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    screen.fill(SPACE)
    pygame.draw.rect(screen, CYAN, (CENTER_X - 20, y, 40, 50))
    # body
    pygame.draw.polygon(screen, WHITE, [(CENTER_X, y - 20), (CENTER_X - 20, y),
(CENTER_X + 20, y)]) # nose
    pygame.draw.rect(screen, RED, (CENTER_X - 16, y + 50, 8, 10))
    # thruster L
    pygame.draw.rect(screen, RED, (CENTER_X + 8, y + 50, 8, 10))
    # thruster R
    pygame.display.flip()
    clock.tick(60)

pygame.quit()
```

Name the three parts you can see and where each one is:

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The body should appear **behind** the nose. Right now the body is drawn last and covers the nose.

```
pygame.draw.polygon(screen, WHITE, [(CENTER_X, y - 20), (CENTER_X - 20, y),  
(CENTER_X + 20, y)])  
pygame.draw.rect(screen, CYAN, (CENTER_X - 20, y, 40, 50))
```

Hint: whatever is drawn last sits on top.

Write the fixed code (show both lines in the right order):

Why was it wrong? Why does your fix work?

Bug B — The nose is supposed to be a **triangle** pointing up, but the point is at the bottom and it looks upside down.

```
pygame.draw.polygon(screen, WHITE, [(CENTER_X, y + 20), (CENTER_X - 20, y),  
(CENTER_X + 20, y)])
```

Hint: to point up, the tip point needs a **smaller** Y than the two base points.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The ship should have **two** thrusters, one on each side. Right now both thrusters land on top of each other in the same spot.

```
pygame.draw.rect(screen, RED, (CENTER_X - 16, y + 50, 8, 10))  
pygame.draw.rect(screen, RED, (CENTER_X - 16, y + 50, 8, 10))
```

Hint: the second thruster needs a different X so it sits on the other side.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Recolour the whole ship with your own palette

At the top of the program, the colours are stored as constants. Change `SPACE` , `CYAN` , and the others to your own colours and run it.

Write the colours you chose and what mood they give your ship:

Add one extra detail to the ship

Add a small detail to your ship — for example a wing line on the side, or a light circle on the nose.

Describe the detail you added and the shape you used to draw it:

Hint: a wing is a `pygame.draw.line` ; a light is a small `pygame.draw.circle` . Remember to draw it **after** the body so it shows on top.

5 · Make It 📷

Build your own three-part ship

Build a scene that has:

1. a body rectangle,
2. a nose triangle on top,
3. two thrusters at the bottom,
4. one extra detail of your own.

Send a **photo or video** of your ship, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I drew the body by ...

I built the nose as a triangle by ...

The draw order matters because ...

My extra detail is ... and I made it with ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough

Take a video of your three-part ship. Talk through it like you are teaching someone. Try to use these words: **polygon**, **triangle**, **palette**, **draw order**, **thruster**.

1. Show all three parts and name them.
2. Read the polygon line out loud and explain its three points.
3. Explain how draw order decides what sits in front.
4. Show the detail you added on top.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your photo or video to teacher on KakaoTalk.